21-242 Recitation 12

Matrices — Q5. Define T A

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Define T_A.

Course-level solution

Definition. For $A \in \mathbb{F}^{m\times n}$, the associated linear map is

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(\vec{x}) := A\vec{x}.$$

Its standard matrix is A itself. Kernel and image of T_A are the nullspace and column space of A.